

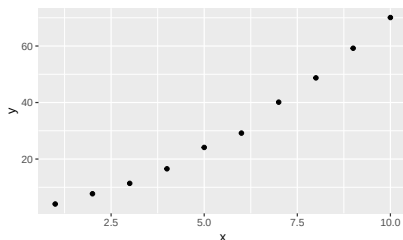
Linear regression and K-nearest neighbors (KNN)

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Overfitting

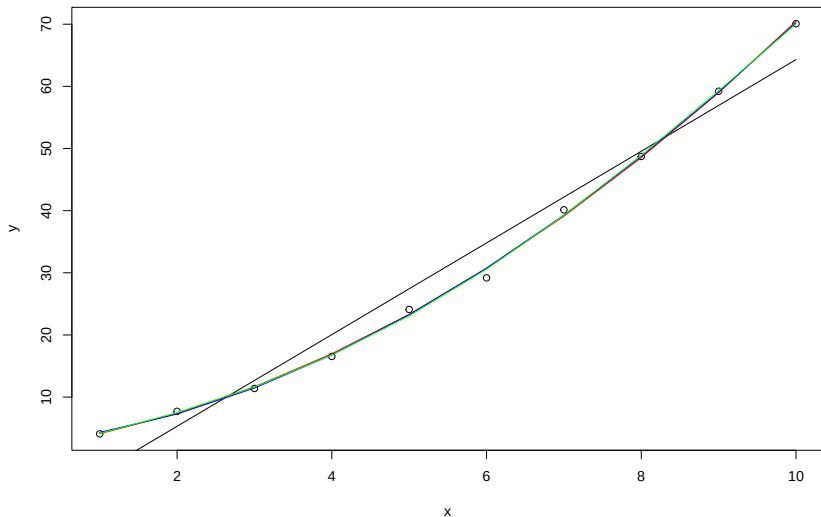
- We revisit the *training* data:

x	y
1	4.09
2	7.71
3	11.39
4	16.55
5	24.11
6	29.18
7	40.13
8	48.72
9	59.22
10	70.08



Polynomial regression

$$Y = \alpha_0 + \alpha_1 X + \alpha_2 X^2 + \dots + \alpha_k X^k + \varepsilon \text{ for } k = 1, 2, 3, 4.$$



Training- and test- MSE



$$trainMSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$



$$testMSE = \frac{1}{m} \sum_{i=n+1}^{n+m} (y_i - \hat{f}(x_i))^2$$

► In our example

$$\hat{f}(x_i) = \alpha_0 + \alpha_1 x_i + \alpha_2 x_i^2 + \dots + \alpha_k x_i^k \text{ for } k = 1, 2, 3, 4.$$

Training-MSE in our example

```
trMSE1=mean((y-predict(reg1))^2)
trMSE1
```

```
## [1] 14.37876
```

```
trMSE2=mean((y-predict(reg2))^2)
trMSE2
```

```
## [1] 0.4588453
```

```
trMSE3=mean((y-predict(reg3))^2)
trMSE3
```

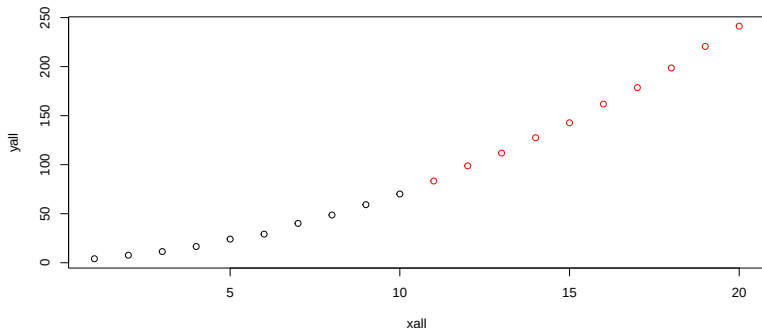
```
## [1] 0.4320397
```

```
trMSE4=mean((y-predict(reg4))^2)
trMSE4
```

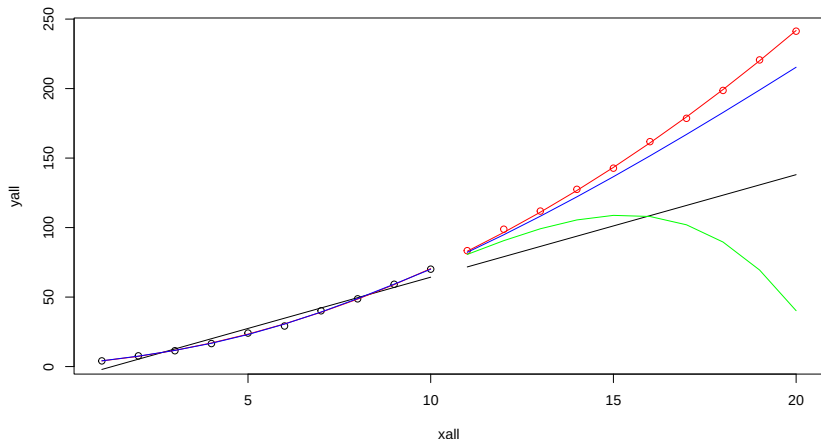
```
## [1] 0.4003628
```

What if we consider other data

- ▶ We actually have 20 observations and we call the last 10 our *test* data. We use this term since we are using this data to test our models.



What if we consider other data



- ▶ The linear model is *underfitting* the data and the 3rd and 4th order polynomials are *overfitting* it. How can we deduce that the data was generated by a 2nd order polynomial?

Test-MSE in our example

```
teMSE1=mean((yout-pred1)^2)
teMSE1
```

```
## [1] 3522.564
```

```
teMSE2=mean((yout-pred2)^2)
teMSE2
```

```
## [1] 0.9843595
```

```
teMSE3=mean((yout-pred3)^2)
teMSE3
```

```
## [1] 173.0981
```

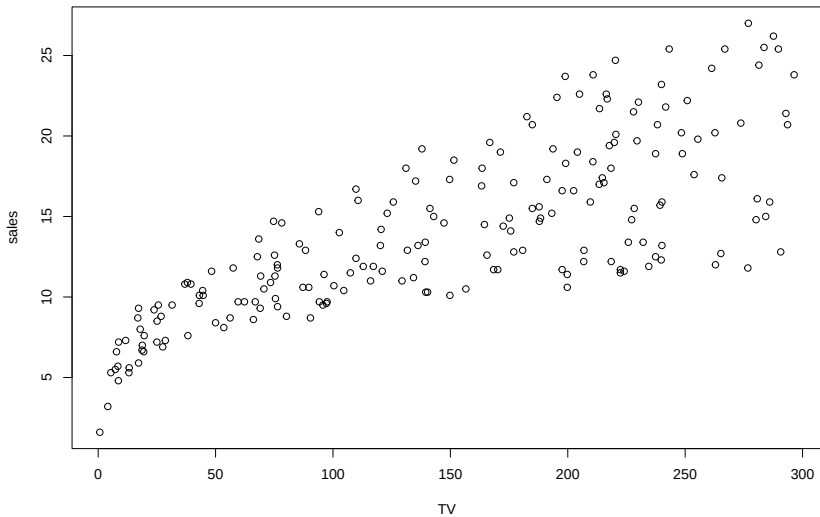
```
teMSE4=mean((yout-pred4)^2)
teMSE4
```

```
## [1] 8582.551
```

Conclusion

- ▶ In the training data, the higher polynomial - the better fit. Among the models we tested, the 4th order polynomial had the lowest training-MSE.
- ▶ In the test data (out-of-sample), overfitting of a model is revealed. The second-order polynomial, which is actually the data generating process (DGP) in this case, is shown to yield the smallest test-MSE.

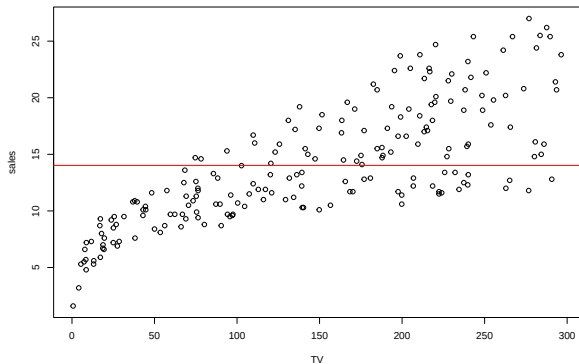
The advertising data



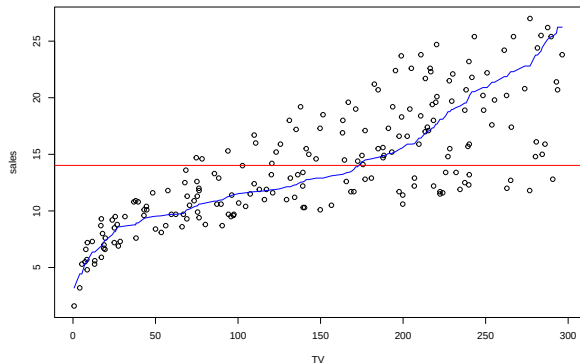
Two extremes

- ▶ Predict with the average value of sales
- ▶ Predict with the average of the sales of the 3 markets which are closest to the markets of interest in terms of TV budget

Average sales over all products



Average of 3 closest products



K-nearest neighbor predictor

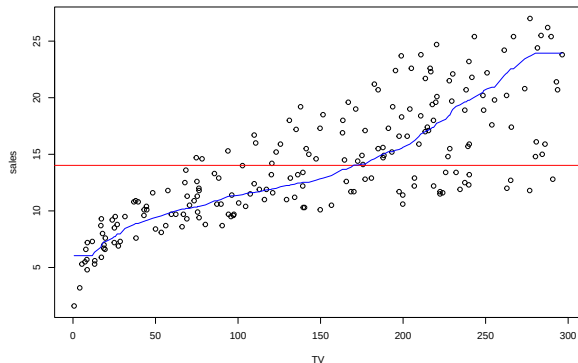
x_0 is a *prediction point* (a test observation)

- ▶ 1. Find the K training-points that are closest to x_0 , call this set $\mathbb{N}_0$.
- ▶ 2. $\hat{f}(x_0) = \frac{1}{K} \sum_{x_i \in \mathbb{N}_0} y_i$
- ▶ With only one predictor x , $\mathbb{N}_0$ contain the K closest observations where “closeness” is measured

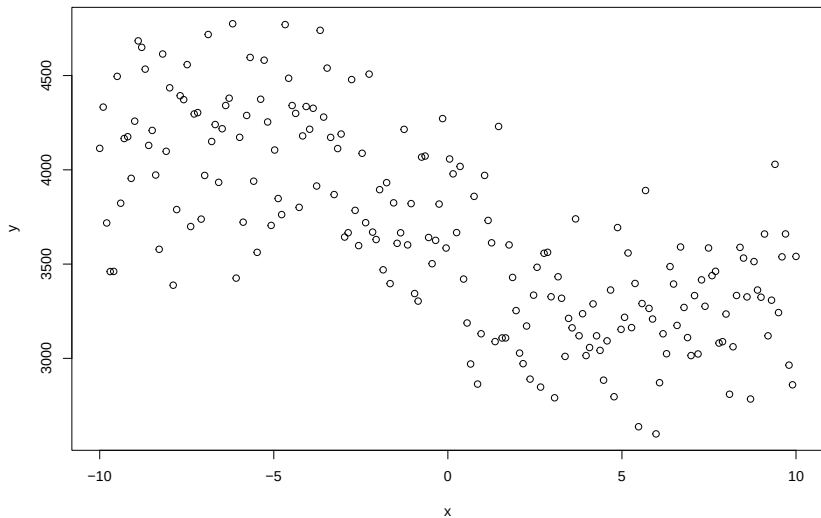
$$d = |x - x_0|$$

Average of 20 closest products

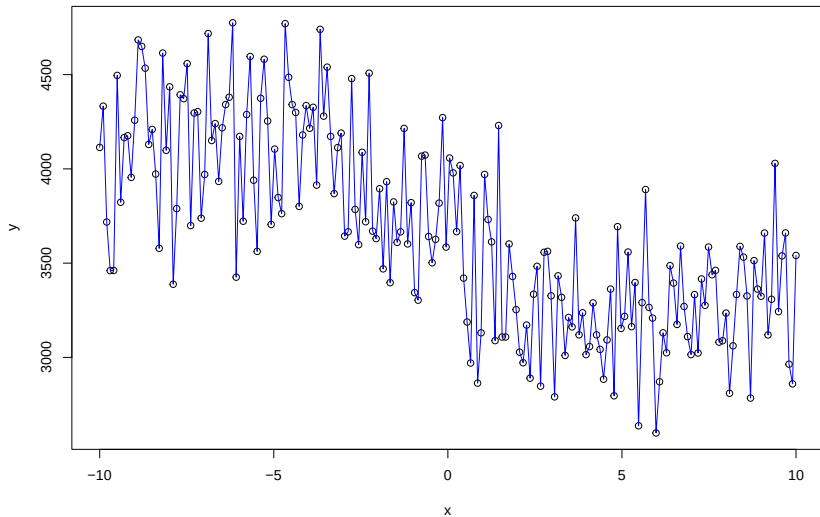
```
knn=function(x0,x,y,K=20)
{
  d=abs(x-x0)
  o=order(d)[1:K]
  ypred=mean(y[o])
  return(ypred)
}
```



Bias-variance tradeoff - exemplifying with KNN

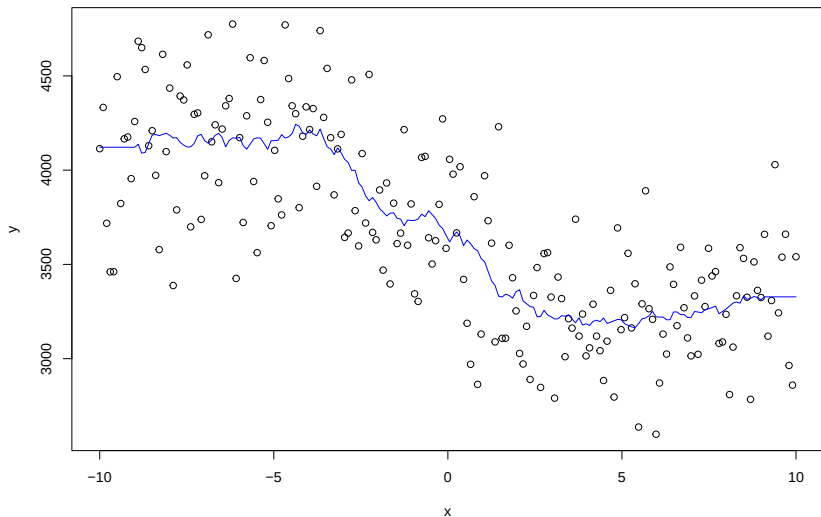


KNN with $K=1$



- A perfect fit in the training data

KNN with $K=20$



- What is the “best” K ?

Bias-variance tradeoff

- Let us first look at the problem of estimating a fixed but unknown parameter θ with the estimator $\hat{\theta}$

$$\begin{aligned}MSE &= E \left((\hat{\theta} - \theta)^2 \right) = E \left((\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2 \right) \\&= E \left((\hat{\theta} - E(\hat{\theta}))^2 \right) + 2 \left(E(\hat{\theta}) - \theta \right) \underbrace{E(\hat{\theta} - E(\hat{\theta}))}_{=0} + \left(E(\hat{\theta} - \theta) \right)^2 \\&= \underbrace{Var(\hat{\theta})}_{\text{Variation}} + \left(\underbrace{Bias(\hat{\theta})}_{\text{Systematic error}} \right)^2\end{aligned}$$

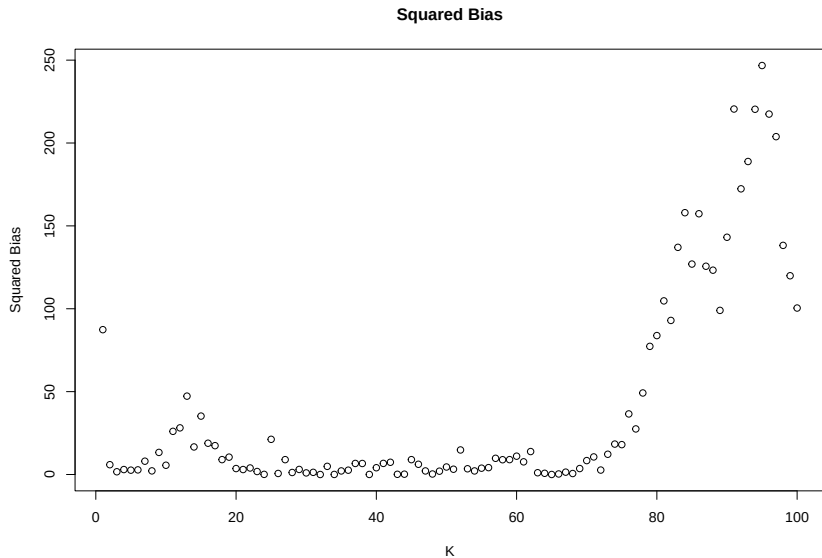
Bias-variance tradeoff

- ▶ For the prediction model based on a predictor X we get an additional component of uncertainty, the irreducible error ε
- ▶ $Y = f(X) + \varepsilon$. Assume we observe a new $X = x$.
- ▶ Mean squared error

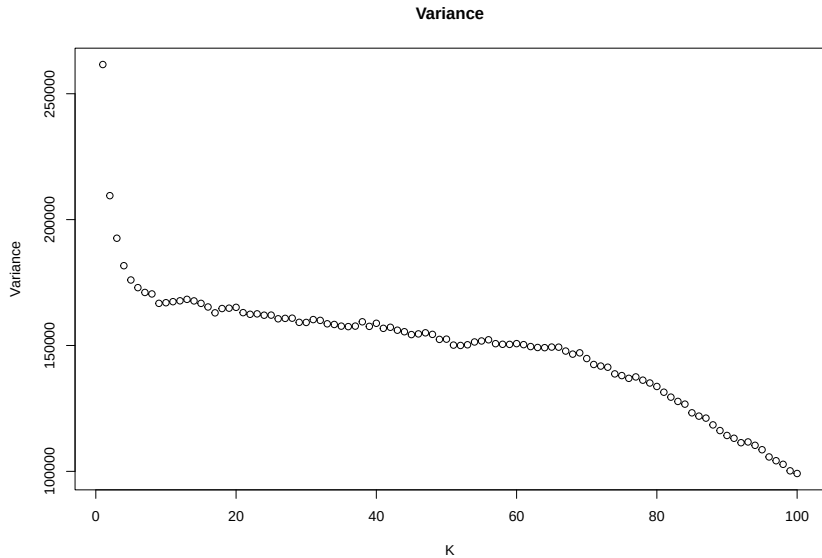
$$\underbrace{MSE(x)}_{\substack{\text{Expected} \\ \text{test MSE}}} = \text{Var}(\hat{f}(x)) + \text{Bias}(\hat{f}(x))^2 + \text{Var}(\varepsilon)$$

- ▶ We search for a $\hat{f}(x)$ with low $MSE(x)$ outside of the sample which we have used to fit the model.
- ▶ In the example where a 4th order polynomial was used to predict, the MSE was very small in the training set (in-sample) but very large in the test set (out-of-sample).

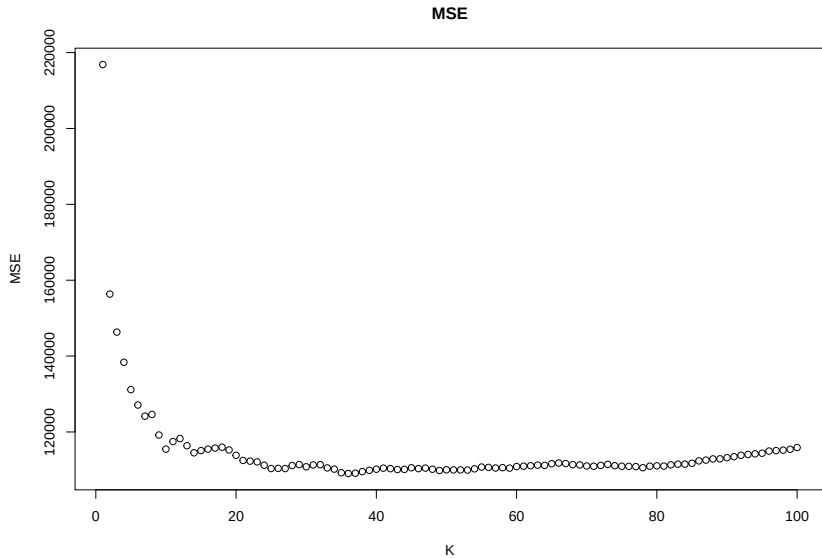
Bias, Variance and MSE for different K



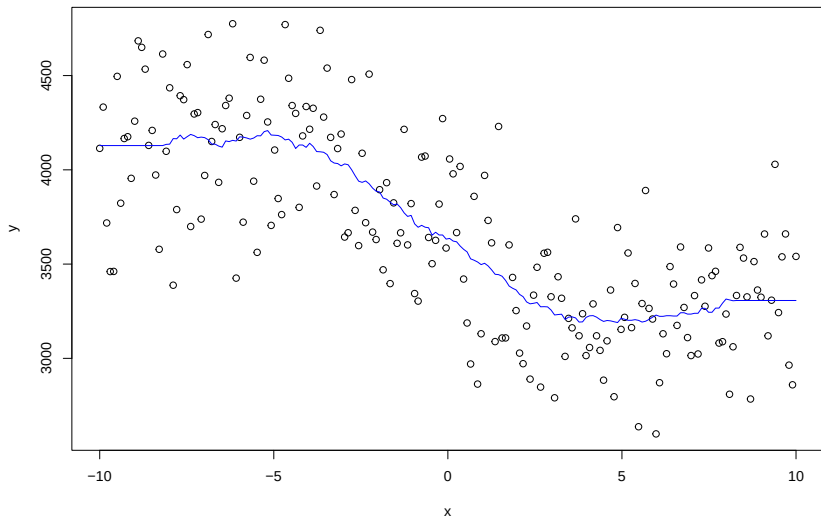
Variance



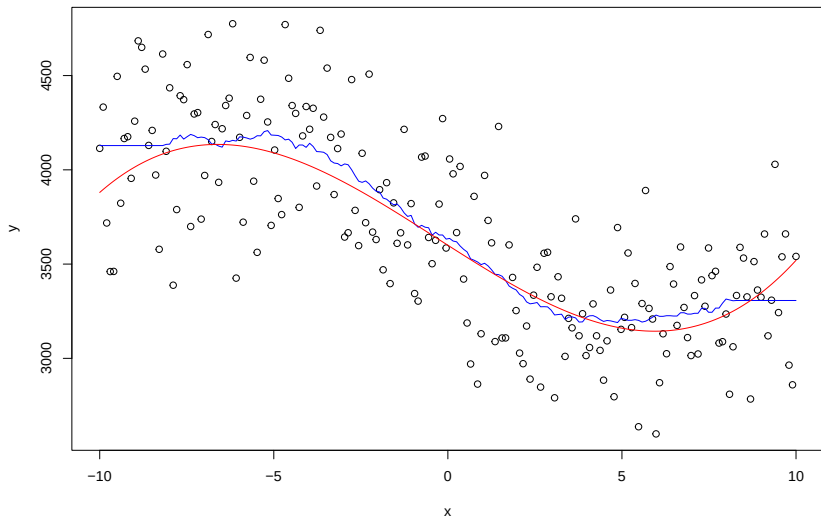
MSE



KNN with “best” $K=36$



Together with the true f



Assessing model accuracy

- ▶ Prediction model $Y = f(X) + \varepsilon$, f unknown, X and ε independent
- ▶ Observations $(x_1, y_1), \dots, (x_n, y_n), (x_{n+1}, y_{n+1}), \dots, (x_{n+m}, y_{n+m})$



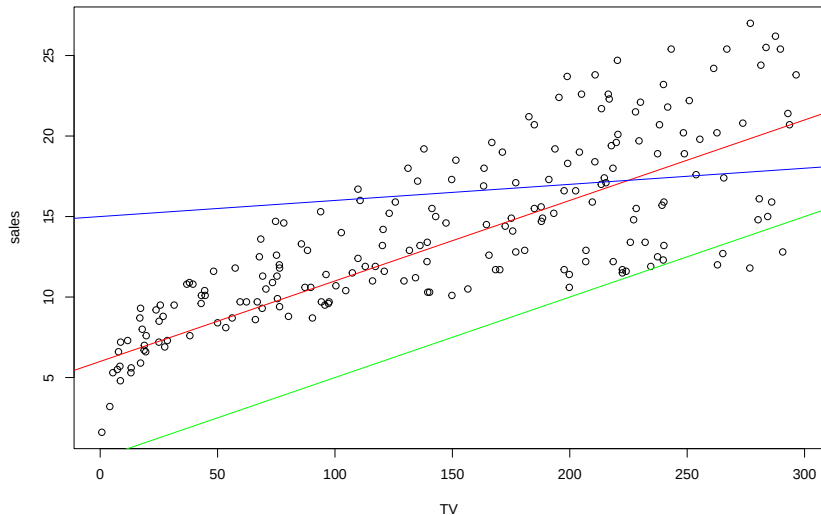
$$\text{trainMSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

measures how observed y_i 's is predicted by $\hat{f}(x_i)$ on average. (Note: MSE is just an example of a measure of fit)

- ▶ $(x_{n+1}, y_{n+1}), \dots, (x_{n+m}, y_{n+m})$ previously unseen (test-) observations not used to estimate (train) $\hat{f}$. We want the method to estimate f with lowest *testMSE*, not lowest *trainMSE*
- ▶ Select the $\hat{f}$ with lowest

$$\text{testMSE} = \frac{1}{m} \sum_{i=n+1}^{n+m} (y_i - \hat{f}(x_i))^2$$

Linear regression - and intermediate solution



Which of the straight lines is most reasonable for prediction of sales?

Linear regression

- ▶ We shall find the straight line $\alpha + \beta x$ that minimizes the sum of squared distances between the line and the points $y_1, y_2, \dots, y_n$. x represents TV ad budget and y represents sales.



$$\underset{\alpha, \beta}{\text{ming}}(\alpha, \beta) = \underbrace{\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2}_{\text{Fitting function}}$$

- ▶ Solved by

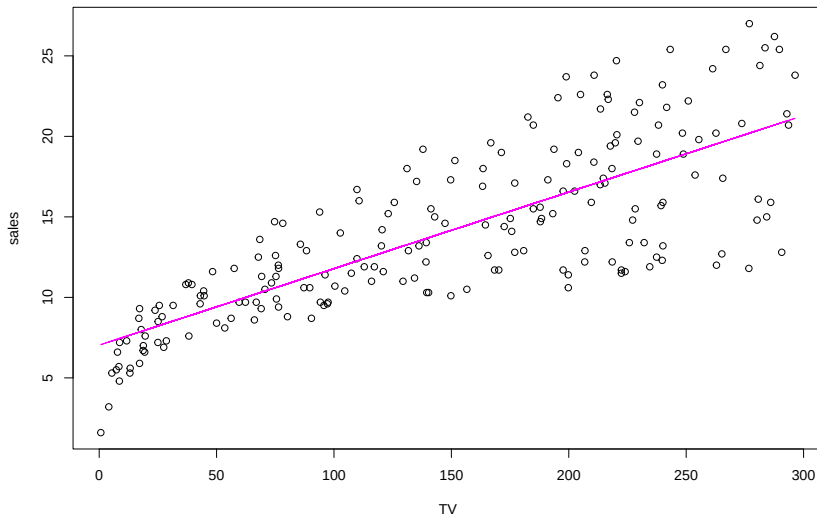
$$\begin{cases} \frac{\partial g(\alpha, \beta)}{\partial \alpha} = -2 \sum_{i=1}^n (y_i - \alpha - \beta x_i) = 0 \\ \frac{\partial g(\alpha, \beta)}{\partial \beta} = -2 \sum_{i=1}^n x_i (y_i - \alpha - \beta x_i) = 0 \end{cases}$$



$$\begin{cases} \hat{\beta} = \frac{\sum x_i y_i - \sum x_i \sum y_i}{\sum x_i^2 - [\sum x_i]^2} \\ \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \end{cases}$$

Linear regression - and intermediate solution

```
reg=lm(sales~TV,data=ad)  
lines(ad$TV,predict(reg),col="magenta")
```



Estimation uncertainty

- ▶ Assume that $y_i = \alpha + \beta x_i + \varepsilon_i$, where ε_i is $N(0, \sigma^2)$ -distributed for all i . In addition, $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ if $i \neq j$.
- ▶ If ε is normally distributed, it follows that

$$\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

Hypothesis testing and confidence intervals for β

If $\varepsilon \sim N(0, \sigma^2)$:



$$\begin{cases} H_0 : \beta = 0 \\ H_1 : \beta \neq 0 \end{cases}$$

If H_0 is true then $z = \hat{\beta} / \sqrt{\sigma^2 / \sum (x_i - \bar{x})^2} \sim N(0, 1)$. If we test on the significance level 0.05, we reject H_0 if $|z| > 1.96$.

- ▶ A 95% confidence interval, *an interval that cover the true value of β with probability 0.95*, is $\hat{\beta} \pm 1.96 \sqrt{\sigma^2 / \sum (x_i - \bar{x})^2}$

Predictions

Model: $Y = f(X) = \beta_0 + \beta_1 X_1 + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$

- ▶ We predict Y by $\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1$
- ▶ A 95% confidence interval for $E(Y|X = x)$, in our example the average *sales* given a particular value of *TV*, is given by

$$\hat{Y} \pm 1.96\sigma \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

Quantifies uncertainty around the average of $Y = \text{sales}$ for a number of markets with (similar) values of $X = \text{TV}$.

- ▶ A 95% prediction interval for is given by

$$\hat{Y} \pm 1.96\sigma \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

Quantifies uncertainty around $Y = \text{sales}$ for one particular market with a given value of $X = \text{TV}$. Accounts for irreducible error.

Multiple regression (using matrix algebra)



$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i,$$

where y_i is a scalar, $\mathbf{x}_i$ a $k \times 1$ vector containing predictor values for individual i and $\boldsymbol{\beta}$ the regression coefficients.

Example

Simple regression with intercept

$$\mathbf{x}_i = \begin{pmatrix} 1 \\ x_{i1} \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\mathbf{x}_i' \boldsymbol{\beta} = \begin{pmatrix} 1 & x_{i1} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha + \beta x_{i1}$$



Ordinary least square estimator - multiple regression



$$\underset{\beta}{\text{ming}}(\beta_0, \beta_1, \dots, \beta_p) = \underbrace{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2}_{\text{Fitting function}}$$



or

$$g(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2$$



is the same as

$$g(\beta) = \sum_{i=1}^n \varepsilon_i^2 = \begin{pmatrix} \varepsilon_1 & \varepsilon_2 & \dots & \varepsilon_n \end{pmatrix} \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} = \boldsymbol{\varepsilon}' \boldsymbol{\varepsilon}$$



or

$$g(\beta) = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

Ordinary least square estimator - multiple regression



$$g(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$$

- ▶ It can be show that this is minimized by the least square estimator

$$\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$$

- ▶ and that the variance-covariance matrix of $\hat{\beta}$ is

$$\sigma^2(X'X)^{-1}$$

- ▶ where σ^2 is the variance of ε_i .

In R

```
reg1=lm(sales~TV,data=ad)
summary(reg1)
```

```
##
## Call:
## lm(formula = sales ~ TV, data = ad)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.3860 -1.9545 -0.1913  2.0671  7.2124
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.032594   0.457843   15.36  <2e-16 ***
## TV           0.047537   0.002691   17.67  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.259 on 198 degrees of freedom
## Multiple R-squared:  0.6119, Adjusted R-squared:  0.6099
## F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

In R

```
X=cbind(1,ad$TV)
head(X)
```

```
##      [,1] [,2]
## [1,]    1 230.1
## [2,]    1  44.5
## [3,]    1  17.2
## [4,]    1 151.5
## [5,]    1 180.8
## [6,]    1   8.7
```

```
y=ad$sales
bhat=solve(t(X)%*%X)%*%t(X)%*%y
bhat
```

```
##      [,1]
## [1,] 7.03259355
## [2,] 0.04753664
```

Accuracy of the model

- Residual standard error:

$$RSE = \hat{\sigma} = \sqrt{\frac{1}{n-2}RSS} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- R^2 : Proportion of variability in Y explained by the model.

$$R^2 = \frac{\overbrace{TSS - RSS}^{\text{not explained}}}{\underbrace{TSS}_{\text{total}}}$$

where

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \text{ and } TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

Overfitting?